Candidate composite value for VAL

Combined factor-model portfolio test

Practitioner factor-model runner

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Abstract

Model. Candidate composite value for VAL, one long-short book from one sleeve (Value). 5857 symbols (full screened universe), 2007-03-01 to 2026-07-01 (233 months).

Earned. 1.37% annual arithmetic return (0.60% compounded), 12.60% volatility, Sharpe 0.11, HAC t=0.42.

Survives? vs SPY market benchmark: alpha 0.80%, t=0.22. vs VAL+MOM+BAB+QMJ: alpha 2.23%, t=1.14. Conventional bar is $|t| \geq 2$.

Best swap. (Standard model, Value for VAL) leaves it unchanged: Sharpe 0.10 against 0.08 (+0.02). **Verdict.** Reject for now: it clears neither control at |t| >= 2.

Exploratory model diagnostic from a single configured run, not evidence from an untouched confirmatory sample.

1 Research objective

Test whether the configured model earns a useful next-month return and whether the benchmark or standard factors explain it. Signals are observed at month-end; returns are next-calendar-month simple returns.

Construction. Normalize within month, combine components within sleeves, then combine sleeves at the stock level using Value 100%. Construct one portfolio; apply beta neutrality and costs once. Sleeve returns are not averaged.

2 Configuration and sample

Factor	Composite signals	Signal transform	Composite transform	Winsor	Min inputs	Mean inputs
Value	book_to_market, earnings_to_price, ps_ttm	rank	rank	1%-99%	2	3.0

Model construction	Book	Beta neutral	Bucket frac.	Cost bps	Min factors	Benchmark
signal weighted	long-short	no	0.33	0.0	1	SPY

Portfolio	Mean stocks	Minimum stocks
Value	791.7	43

Sample: 5,857 symbols, 195,258 stock-months, 2001-10-01–2026-08-01. Screens: price 1.0, 21-day dollar volume None, market cap None, top-cap fraction 0.375, exclude financials True. Benchmark: SPY. Factor controls: standard_recreated_factors.csv.

3 Portfolio returns

Raw net performance on one shared window. Return and volatility use monthly arithmetic moments; t is HAC.

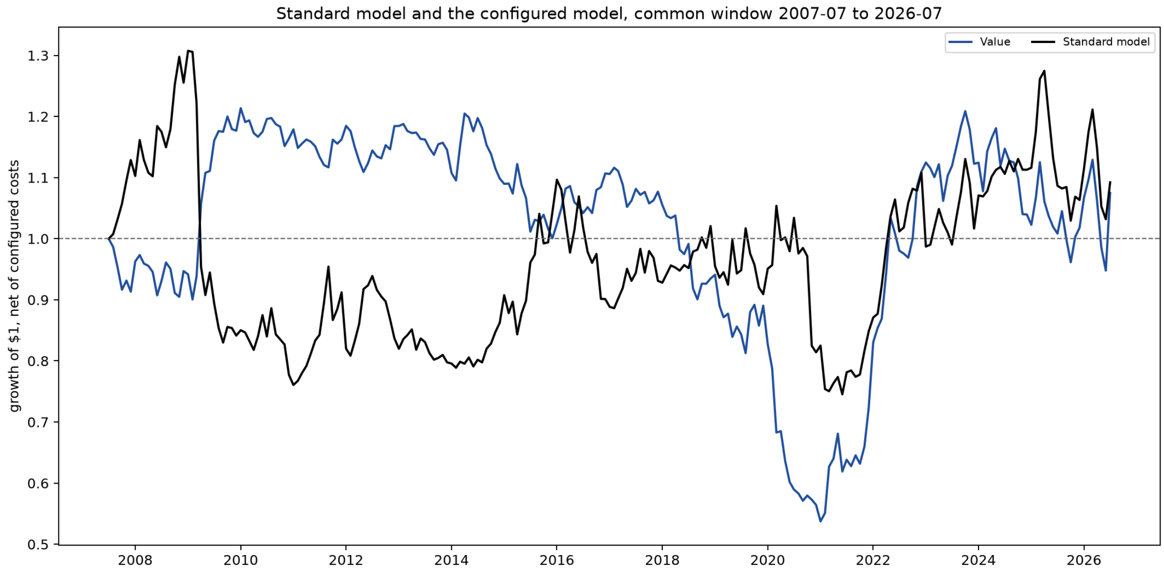

Figure 1: Growth of one dollar in each book, net of configured costs, on the common window. The path begins at one before the first return month.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Value	233	1.37%	0.60%	12.60%	0.11	0.42
Standard model	228	1.49%	0.47%	14.16%	0.11	0.45

4 Benchmark fit and benchmark-orthogonalized returns

Regress each raw net return on SPY market benchmark. R2 is variance explained; alpha is the annualized intercept; t and F use HAC inference. This is separate from standard-factor attribution. Estimated MKT loading: 0.049. Annual raw return 1.37% = controlled alpha 0.80% + benchmark contribution 0.57%. Orthogonal means zero sample covariance with the benchmark, not a lower mean. This is an ex-post regression diagnostic, not beta-neutral construction or a tradeable improvement.

Portfolio	N	Loading (MKT)	t (loading)	R2 (SPY market benchmark)	F
Value	233	0.05	0.60	0.004	0.36
Standard model	228	-0.56	-9.19	0.376	84.50

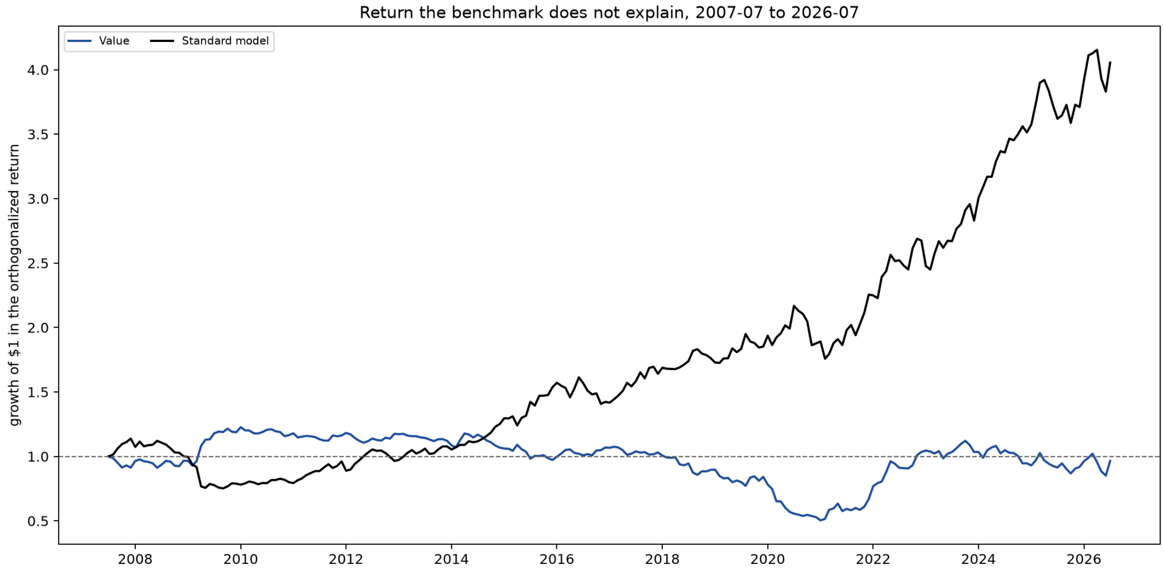

Figure 2: Returns orthogonalized to the SPY market benchmark: growth of one dollar in each book once the SPY market benchmark component is removed. A flat path earns nothing the benchmark does not already deliver.

The table summarizes the residual paths above.

Portfolio	**Months**	**Ann. ret.**	**CAGR**	**Ann. vol.**	**Sharpe**	**t**
Value	233	0.80%	0.03%	12.58%	0.06	0.24
Standard model	228	8.02%	7.65%	11.18%	0.72	3.04

5 Attribution to configured factor controls

Regress each raw net return simultaneously on VAL+MOM+BAB+QMJ: $r_{p,t} = \alpha + \sum_k \beta_k f_{k,t} + \varepsilon_t$. The first table gives the loadings, with HAC t in brackets.

Portfolio	**VAL**	**MOM**	**BAB**	**QMJ**
Value	0.62 (7.20)	-0.33 (-5.88)	0.08 (1.29)	0.06 (0.87)

Return contribution is $\beta_k \times 12 \times \overline{f_k}$. Explained plus alpha equals total annual return.

Portfolio	**VAL**	**MOM**	**BAB**	**QMJ**	**Explained**	**Total**
Value	-0.67%	-1.84%	0.24%	0.50%	-1.78%	0.45%

5.1 Simultaneous factor spanning and factor-orthogonalized returns

The same regression as a spanning test. R2 is the share of monthly net-return variance jointly explained by the factor controls.

Portfolio	**N**	**R2 (4F)**	**F**
Value	230	0.642	56.44

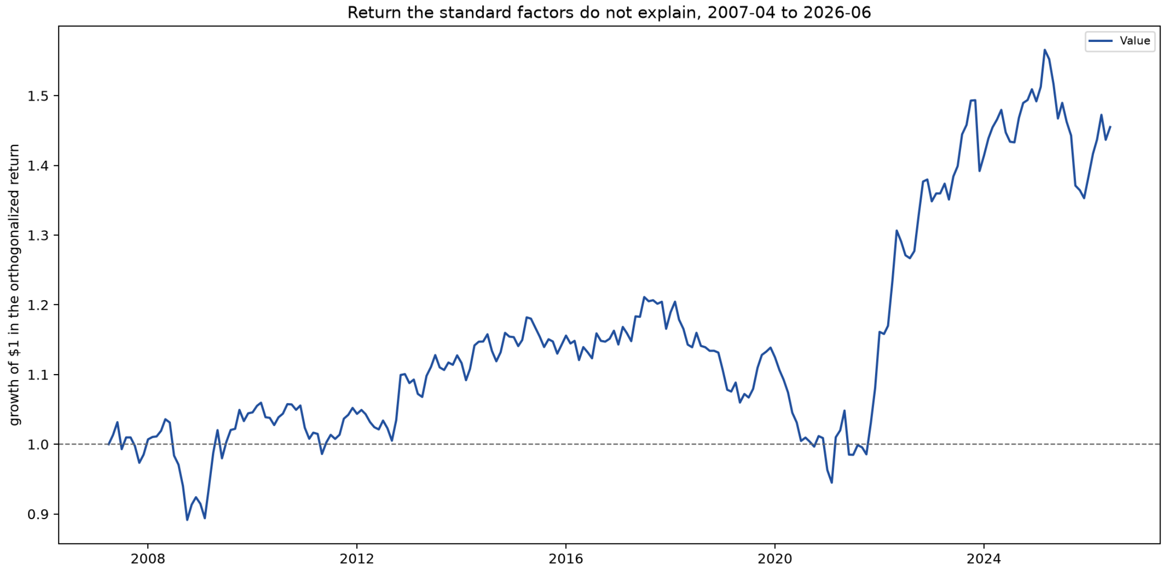

Figure 3: Returns orthogonalized to VAL+MOM+BAB+QMJ: growth of one dollar in each book once its factor exposures are removed. This is the Alpha column above, drawn as a path.

The table summarizes the factor-residual paths above.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Value	230	2.23%	1.98%	7.35%	0.30	1.26

5.2 The new model against the reference model

Regress the combined model on Standard model as one return series. High R2 with zero alpha means the reference reproduces the model.

Portfolio	N	Alpha	t	R2 (Standard model)	F
Value	228	1.29%	0.39	0.008	0.46

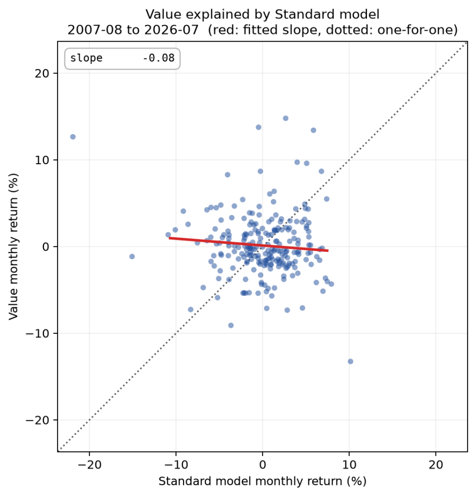

Figure 4: Each month of the combined model against Standard model, with the fitted slope. A tight cloud along the line is a model the reference reproduces; vertical spread is what it cannot.

5.3 Correlation and principal components

Check whether a sleeve duplicates a standard factor. The combined model is excluded because its overlap with its own sleeves is mechanical. PCA uses standardized sleeve and standard-factor returns over 230 common months. The share shown in the plot is the largest covariance-matrix eigenvalue divided by the sum of all eigenvalues: the fraction captured by one dominant co-movement, not return or regression R2. The combined model is excluded because its overlap with its own sleeves is mechanical.

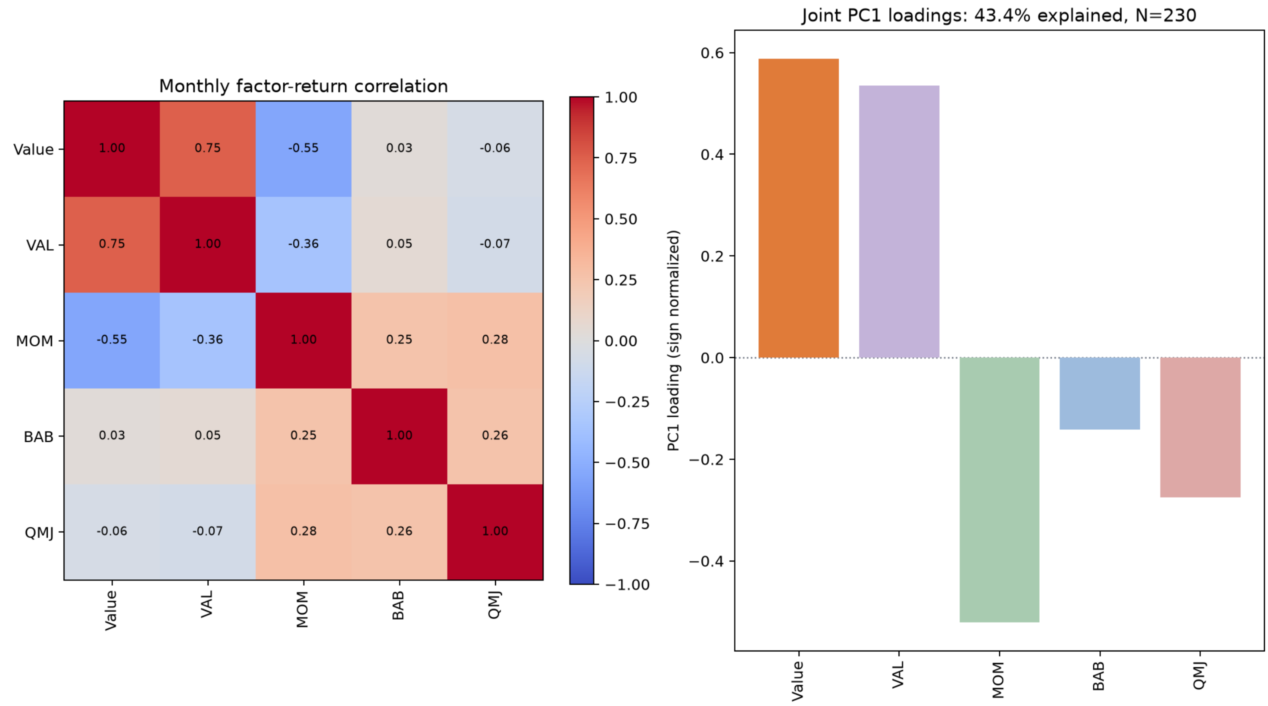

Figure 5: Monthly return correlations and sign-normalized first-principal-component loadings across the standalone sleeves and the supplied standard factors.

6 Signal persistence

Mean rank IC between formation exposure and a single later month's return. Flat profiles persist; fast decay implies faster rebalancing.

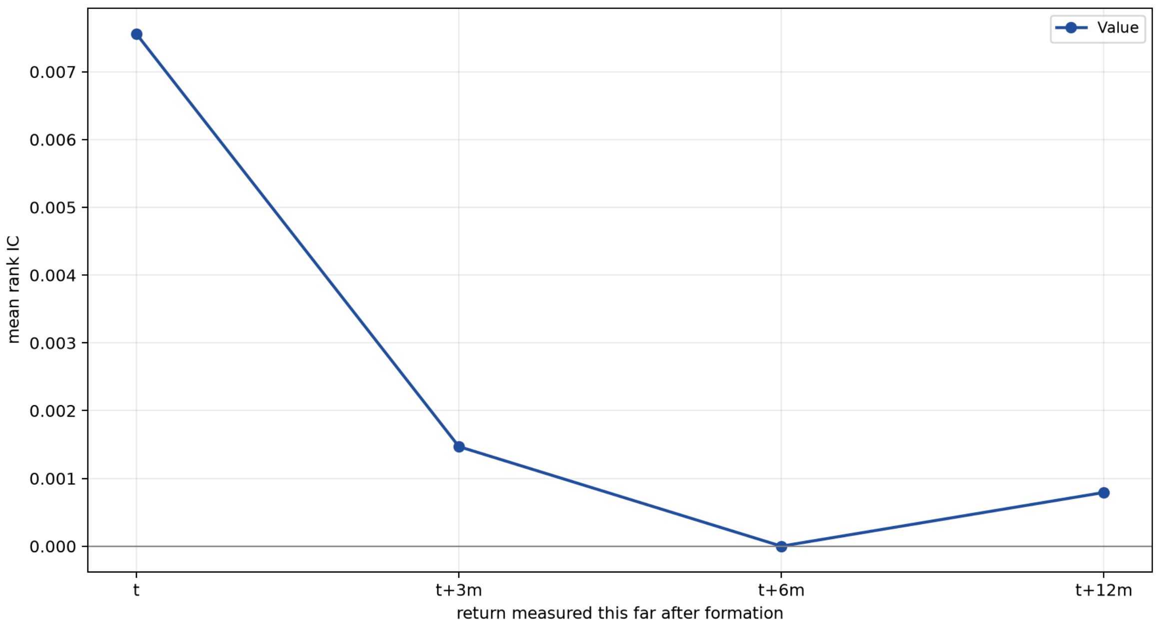

Figure 6: Mean rank information coefficient at each horizon. A line that stays flat is a signal that keeps working long after the month it is traded.

Portfolio	Horizon	Mean IC	t	Months
Value	t	0.0076	0.70	233
Value	t+3m	0.0015	0.13	230
Value	t+6m	-0.0000	-0.00	227
Value	t+12m	0.0008	0.07	221

7 Candidate comparisons and replacement

Standalone compares candidate with incumbent. In-model replaces the incumbent sleeve and rebuilds the standard model; diversification can reverse the standalone ranking.

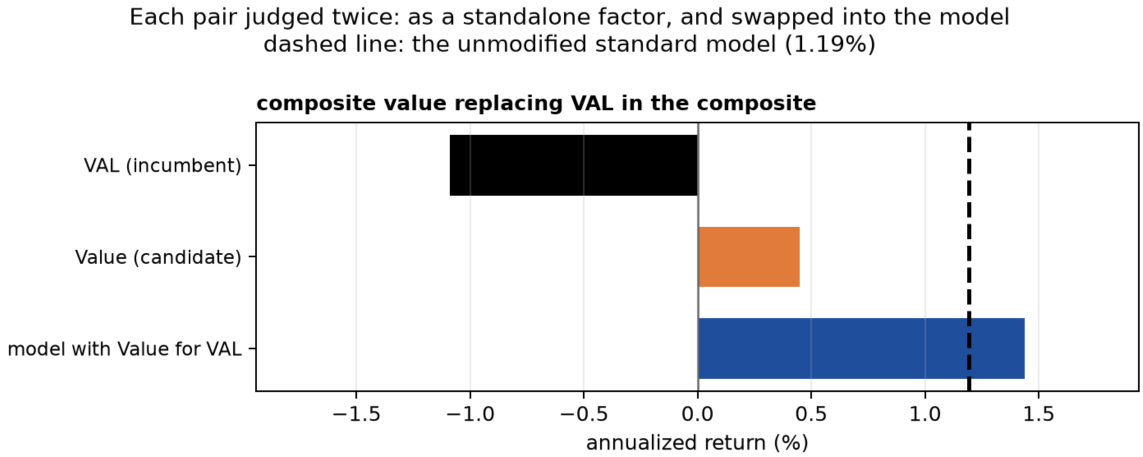

Figure 7: Each pair judged twice: the incumbent factor against the candidate standing alone, then the model with the candidate swapped in. The dashed line is the unmodified standard model.

Pair	Role	Portfolio	Ann. ret.	Change	Alpha	Change	t
composite value replacing VAL in the composite	incumbent	VAL	-1.09%	–	-0.77%	–	-0.23
	candidate	Value	0.45%	+1.54%	-0.13%	+0.64%	-0.04
	in model	Standard model, Value for VAL	1.44%	+0.25%	7.80%	+0.07%	3.04

8 Conclusion

Verdict. Reject for now: it clears neither control at |t| >= 2. vs SPY market benchmark: alpha 0.80%, t=0.22; vs VAL+MOM+BAB+QMJ: alpha 2.23%, t=1.14.

9 Reproducibility

Run ID:

`20260826T135907Z_candidate-composite-value-for-val_79b798972d`

Configuration hash: `79b798972d`. Gold version: `version=20260826T131322Z`. The run directory contains configuration, data products, figures, source, log, and PDF. Re-runs never overwrite it.